

Jonathan Ye

Stanford, CA

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EDUCATION

Stanford University

B.S. in Computer Science, GPA: 3.96

Stanford, CA

Sep. 2023 – June 2027

EXPERIENCE

Amazon

June 2025 – Sep. 2025

Software Engineer Intern

Dallas, TX

- Built an AI agent that analyzes Cloudwatch logs across accounts using Python, AWS, React.js, and Strands
- Improved response times by 400% by migrating the Lambda service from Docker image to Amazon Peru deployment and reducing build size from 1 GB to 200 MB
- Provided internal access to the agent by setting up an API and calling it with a React.js UI as a use case
- Streamlined Lambda service setup and cross-account Cloudwatch access using CloudFormation and IAM Roles

Stanford Cybersecurity and AI Lab

Dec. 2024 – June 2025

Researcher

Stanford, CA

- Enhanced Cybench and Bountybench's functionality—cybersecurity benchmarking tools for LLM models—by developing and integrating high-quality CTF challenge and bug bounty tasks into the CI using Python and Docker
- Interacted with server-side software through the configuration and testing of API requests and endpoints
- Improved user productivity by 300% by adding features on front-end and agent workflow on the back-end
- Safely implemented changes by developing unit tests and collaborating with reviewers to ensure code integrity

Grade Soar

July 2024 – Sep. 2024

Software Engineer Intern

Remote

- Designed and deployed a robust learning management system through effective team collaboration
- Delivered a responsive user interface by leveraging the MUI component library and Flexbox
- Enabled parameter customizability in components using Javascript

PROJECTS

Poke Pocket Market | Javascript/Typescript, React.js, Next.js, Firebase, Digital Ocean

- Built and deployed a trading platform for 100+ active users who play Pokemon TCG Pocket
- Designed Firestore database and algorithms that improve operation efficiency by 500% on user and card records
- Delivered a modern user interface and user experience with Tailwind CSS and ShadCN components
- Auto-update cards by data scraping 1000 cards/second with Python and storing data in Firebase Cloud Storage
- Allowed HTTPS connection using Nginx reverse proxy to redirect traffic within Digital Ocean server

Aspen | Discord.py, OpenAI API, LangGraph, Exa API, Python

- Built an AI-powered therapist that monitors Discord channels and reaches out to individuals in distress
- Directed the flow of conversation phases with LangGraph and OpenAI prompting
- Implemented web searching/crawling with Exa API to use intake information and design a therapy plan

Data Processor and Map Dashboard | Javascript/Typescript, React.js, Next.js API, Mapbox API

- Retrieves 200 normalized housing listings/second by processing and cleaning raw API data with Typescript
- Improved data fetching speed by 1000% with optimized API route interaction with PostgreSQL and Prisma
- Displays and imports 1000 listings/second into a customized Mapbox API component

Project Launchpad | Unity, C#, FL Studio, Audacity

- Created and released a Guitar Hero/Osu-inspired rhythm game utilizing all letter keys for note interactions
- Synchronized visual cues with note interactions using Unity's particle system for enhanced gameplay
- Streamlined creating and importing levels stored in JSON files through Unity's Data Persistent Path

TECHNICAL SKILLS

Languages: Python, Javascript/Typescript, C, C++, Java, C#, HTML/CSS

Frameworks: React.js, Next.js, Firebase (NoSQL), Unity

Technologies: Git/Github, AWS, Docker, Linux, Digital Ocean, Nginx

Concepts: Full-Stack, GenAI Integration, Operating Systems, Data Structures & Algorithms